

Software Architecture Specification

ENGINEERING • SYSTEM DESIGN & MICROSERVICES

1. Microservices Breakdown

The AI Knowledge OS consists of distinct microservices: Document Parser Service, Embedding Ingestion Worker, FAISS Vector Engine Service, and Query Orchestrator API.

2. Data Flow Architecture

Documents -> Preprocessing Engine -> Dense Embedding Generator -> FAISS Vector Database -> Context Assembler -> LLM Interface.

3. Scalability & Latency

Target P99 query latency is under 250ms. Vector search nodes scale horizontally behind a load balancer with sharded memory tables.